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International AS and A-level

Physics (9630)

PH04: Energy and energy resources. Report on the Examination

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REPORT ON THE EXAMINATION: INTERNATIONAL AS/A-LEVEL PHYSICS (9630) PH04 – JUNE 2026

This paper was similar to those of previous years in terms of specification coverage and demand. However, there was a increase in the mean mark of around 3 marks compare to Summer 2025. Students had many opportunities to demonstrate their knowledge, skills and understanding across a range of topics. The usual variety of question styles including short answers, single and multi-step calculations, extended writing and multiple choice were used. The questions were set in a range of both familiar and unfamiliar contexts.

The balance of assessment objectives and the mathematical demand was similar to those of previous series.

The best answered questions tended to be straightforward calculations. Most students were also able to cope with the many multi-step calculations on the paper. Many students had much more difficulty with extended writing and explanations.

SECTION A

QUESTION 01

This question assessed the ability of students to apply their knowledge and understanding of thermal physics to a type of water heater that uses wood. All four questions parts were calculations and had a good level of discrimination.

01.1 was answered correctly by about 65% of students. Most of those who did not get full marks had difficulties converting 11 kW h into J. A very small minority of students made a mistake with algebra, dividing 19.2 MJ by their number of J. Most students realised that the answer had to be to 2 sf.

01.2 used a cognitive step mark scheme that rewarded students who did not get the correct answer but set out their work in a clear way so that intermediate credit could be awarded. The question behaved in a similar way to 01.1, with 63% getting all four marks and 93% getting at least 3 marks. Despite this, the questions discriminated very well. The most common error was to give the answer to just one significant figure, giving an answer of 0.2 (kg). An acceptable answer would have been 0.20 (kg). The mark scheme also contained the calculator value which meant that examiners checked the final answer and penalised incorrect rounding. Another relatively common error was with the third bullet point, the correct use of the efficiency of the system. Some students simply missed this out.

01.3 was also done very well by about 60% of students. Again a cognitive step mark scheme was used. Those who did not get full marks commonly assumed that the temperature difference was negligble as part of their calculation. Others assumed the insulation, rather than the wall, was on the hot side of the system. The mark scheme meant that these students could get partial credit provided their working was clear. There were several valid routes for this question, all of which were given credit. This was one of the most discriminating questions on the paper.

01.4 required students to carry out a relatively straight-forward calculation and about 80% of students obtained both marks. Although it was rare, the use of Boltzmann's constant for k was seen. This suggests that these students had very little understanding of this topic. The discrimination for this questions was also very high.

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QUESTION 02

This question assessed students' knowledge and understanding of gasses in the context of a helium party balloon. All three questions were calculations involving several steps and all discriminated well.

02.1 84% of students obtained all four marks for this multi-step calculation. Those who did not often made an error converting the temperature to K or missed out Avogadro's number after using pV = NkT.

02.2 This was a much more straight-forward calculation and that was again completed correctly by about 84% of students. Common errors included using the diameter rather than radius or using the incorrect formula for the volume of the sphere. Some students were not given MP2 because what they calculated in MP1 was not a volume.

02.3 This question was slightly more demanding with about 74% of the students getting full marks. Its discrimination was one of the best on the paper. A very common error was the use of the wrong mass for the helium atom.

QUESTION 03

This question was about the closest approach of alpha particles to a lead nucleus.

03.1 This question discriminated very well. About 77% of students obtained all four marks. The mark scheme rewarded students who did not get full marks but set out their work clearly so that credit worthy work could be seen. Common errors included missing out the alpha particle charge, using the force rather than energy equation and using the proton number rather than the nucleon number in the equation for R_o.

03.2 This was the first question that required extended writing and was answered fully by less than 20% of students. The discrimination was also relatively low compared to the previous questions. Many students failed to get a mark for simply referring to 'upper limit' without explaining what that means. Many also gave an answer in general terms rather than taking into account the data in the equation. For example some suggested that the answer did not take into account the strong force despite calculating a value for the radius which was clearly much greater than the range for that force. Very many answers had insufficent detail for marks to be awarded. About one third of students obtained zero on this question.

QUESTION 04

This question was about nuclear fission.

04.1 This was a six mark level of response question that produced a good spread of marks and discriminated well. Most students understood the role of the moderator in a fission reactor and could give some account of why it is required. The description of moderation was generally well done but some students incorrectly referred to neutron collisions with atoms or molecules rather than nuclei. The best answers described a system of collisions that gradually reduced the speed of the neutrons so that they were in thermal equilibrium with the moderator. Most students were able to give one correct example of a moderator. Answers that explained the importance of the low nuclear mass were less common. More than 12% of students obtained full marks. About 12% also obtained zero marks, including 5% who did not attempt the question.

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04.2 68% of students obtained full marks for this question and it discriminated very well. Common errors included attempting to include the mass-energy of the neutrons in their answer or getting the equation for the released energy wrong. Some students did not convert their binding energies per nucleon into binding energies. If they obtained an energy, most students were able to convert this to a mass correctly.

04.3 This was one of the most difficult questions on the paper with only 9% of students obtaining all three marks. About 60% obtained zero, including 25% of students who made no attempt. Some students did not make progress because they referred to binding energy rather than binding energy per nucleon. Most answers were either vague or incomplete without stating much more than the information given in the question.

QUESTION 05

This question was about rotational motion in the context of rolling cylinders down a slope. There were seven question parts that gradually took the students through the ideas behind rotational motion.

05.1 This was one of the most straight-forward questions on the paper. Issues were seen with how some students formed their letters, particularly the I for the moment of inertia. Students should be encouraged to take much more care when writing the symbols that refer to physical quantities.

05.2 This question required students to describe an experiment to determine a time for a cylinder to roll down a slope. Many answers gave little idea of what to measure, how to measure it or what to do to make the value accurate. Students should be encouraged to use these points when writing an experimental method. Only about 44% of students obtained full marks with about 19% obtaining zero. Of these, about 5% made no attempt.

05.3 This was the most difficult question on the paper with only 7% getting both marks. Most students were able to suggest that the cylinder would not roll, but explanations were often incomplete. The best answers stated clearly that the remaining forces would act through the centre of mass and that there would therefore be no turning effect.

05.4 This was much more accessible, with 84% of students getting both marks. Some students used the incorrect formula for the circumference or made arithmetic errors in their calculation. The discrimination was very high.

05.5. This calculation was completed correctly by 86% of students. A common error was using the wrong trigonometrical function.

05.6. This was the most discriminating question on the paper. This was because there were four very distinct steps in the calculation, each of which was marked independently. About 76% of students completed this calculation correctly. The most common mistake was to fail to take into account the translational kinetic energy of the cylinder.

05.7 In this question students were required to deduce which cylinder would reach the bottom of a slope first. It proved to be very demanding with only about 12% of students getting all three marks. About one third of students obtained zero, including 5% who made no attempt. Whilst many students stated that X would have the greater moment of inertia, very few explained why. The best answers went on to take an energy approach explaining clearly why Y would therefore have the greater translational kinetic energy and reach the bottom first. Students should be encouraged to write out answers to questions of this kind logically and clearly, explaining each step in their argument.

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QUESTION 06

The question was about alternative energy sources.

06.1 This question was a 'show that'. This means that students were required to set out each step of their algebra clearly. About 56% of students were able to do this and obtain full marks. The most common error was to miss out a step, such as the density equation, or to treat V as the volume per second without explanation. Many answers were also seen in which the v for speed and V for volume were indistinguishable. Students should be encouraged to make their symbols much clearer and to explain what they are doing in questions of this kind.

06.2 This question was much more accessible with 75% of students completing the calculation correctly. There was some confusion between diameter and radius and power of ten errors were also seen. Some very successful answers were seen that derived the equation for mass per second and gave the correct value.

06.3 About 45% obtained full marks for this graph. This was partly because the mark scheme allowed one error and still gave full marks. Students often lost marks for careless lines. They should be encouraged to use a ruler when a line is straight. Curves often went through incorrect points or had flat peaks. The question produced a good spread of marks.

SECTION B

The mean mark on this section was 84%. The students found none of the questions particularly difficult.

The most straightforward, with a mean mark greater than 85%, were B7, B9, B20, B15, B21, B16 and B18.

The most difficult was B14, which was about the hydrogen cycle in the Sun. 68% of students answered it correctly. The most popular distractor was B. It may be that these students forgot the two extra hydrogen nuclei that enter the cycle at the second stage.

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